

SAFETY OPERATING PROCEDURES

Milling Machine

DO NOT use this lathe unless you received training by a certified instructor and you have permission to use the equipment.



Safety glasses must be worn at all times in work areas.



Sturdy footwear must be worn at all times in work areas.



Rings and jewellery must not be worn.



Gloves must not be worn when operating the machine.



Long and loose hair must be contained.



Close fitting/protective clothing must be worn.



Hearing protection may be required for some operations.



Cell phones must be turned off at all times in the machine shop.

IT IS EXPECTED THAT THESE PROCEDURES BE FOLLOWED AND MAINTAINED IN ALL INSTANCES
Should an incident occur it is imperative it is reported to your instructor or call 9-911.
Please leave all as it is so a review can take place.

PRE-OPERATIONAL SAFETY CHECKS

1. Ensure no slip/trip hazards are present in workspaces and walkways.
2. Locate and ensure you are familiar with the operation of the ON/OFF starter and E-Stop.
3. Do not leave equipment on top of the machine.
4. Check that machine guards are in position.
5. Ensure cutter is in good condition and securely mounted.
6. Check coolant delivery system to allow for sufficient flow of coolant (if needed).
7. Faulty equipment must not be used. Immediately report suspect machinery.

OPERATIONAL SAFETY CHECKS

1. Keep clear of moving machine parts.
2. Never leave the machine running unattended.
3. Follow correct clamping procedures- keep overhangs as small as possible and check work piece is secure.
4. Set the correct speed to suit the cutter diameter, the depth of cut and the material.
5. Before making adjustments and measurements or before cleaning swarf accumulations switch off and bring the machine to a complete standstill.

HOUSEKEEPING

1. Switch off the machine.
2. Remove milling cutters and store them safely.
3. Leave the machine and work area in a safe, clean and tidy state.

POTENTIAL HAZARDS

- Sharp cutters
- Moving components - hair/clothing entanglement
- Eye injury
- Skin irritation
- Metal splinters and burrs

This SOP does not necessarily cover all possible hazards associated with the machine and should be used in conjunction with other references.
It is designed to be used as an adjunct to teaching Safety Procedures and to act as a reminder to users prior to machine use.

- Flying debris

FORBIDDEN

- Cell Phones must be turned off at all times in the machine shop.

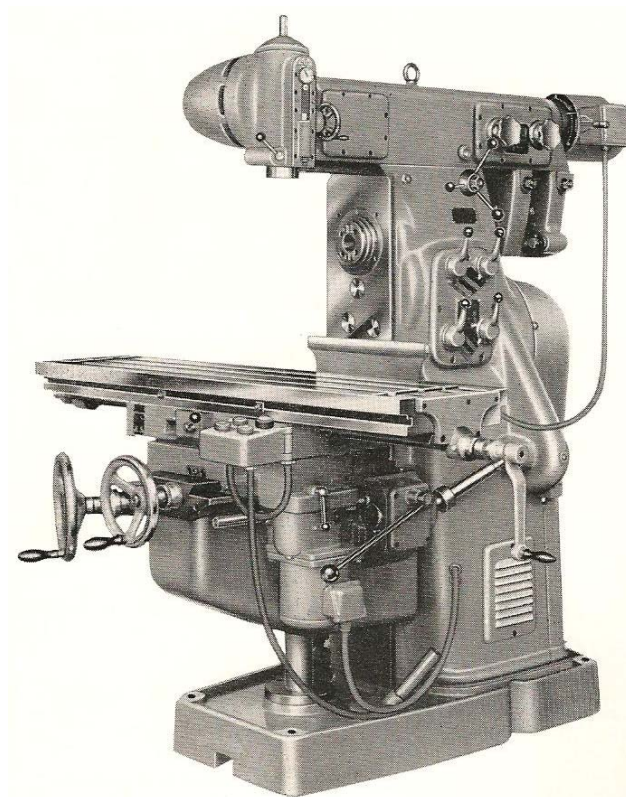


Figure 1 – Pedersen PU2 SV13 Milling Machine

General principles concerning operator safety for milling machines

1. Do not grip a workpiece with grease or oil on it.

Grip all workpieces firmly.

Do not attempt to hold workpieces that are too awkward or too difficult to hold.

Do not hold workpieces that are too heavy for the machine.

Know how to hold workpieces properly when lifting.

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2. Be sure to clean oil or grease from hand tools, levers and handles.

Be sure there is enough texture on the surface of the hand tool or lever handle for proper safe hand contact.

3. Grip hand tools and lever handles firmly.

Always choose the proper hand tool and appropriate grip position on the lever handle.

Do not use hand tools or lever handles in an awkward position.

Do not apply excessive force.

4. Always use the recommended gripping position to grasp hand tools and lever handles.

5. Do not allow tools to be caught in the milling cutter or other holding device.

6. Do not use broken, chipped or defective tools.

7. Be sure the workpiece is securely held in the vice, chuck or other holding device.

8. Beware of irregular shaped workpieces.

9. Beware of large burrs on workpieces.

10. Always select the correct tool for the job.

11. Do not run the machine unattended.

12. Do not use tools without handles.

13. Always support the workpiece as necessary.

14. Correctly locate tool in socket heads and screw slots.

15. Beware of obstructions that prevent complete tightening of screws – ensure screw is tight.

16. Do not rush work.

17. Never substitute the wrong size tools if the correct sized tool is not available or cannot be located in the shop.

18. Do not place hand or body in path of moving objects.

Beware of moving milling parts that can fall.

Be aware of where you are moving your hand or body in relationship to the milling machine.

Be aware of hands or other parts of the body that may in position to be hit by a cutting tool or workpiece.

19. Beware of accidentally moving levers, clutches (when applicable) or turning the power on.

20. Know the function of each and every control.

21. Never place a hand on the tool to stop the rotation of the spindle.

22. On machines with clutch drive make sure the clutch is completely disengaged on stopping, and kept properly adjusted.
23. Make sure the power has been turned off when the milling machine is unused for sometime.
24. Allow the spindle to stop before operating it.
25. Always check the spindle area for tools and loose items.
26. Do not allow distractions to interfere with milling machine operations.

Do not operate a milling machine whilst talking.

27. Beware of milling machine dangers when attending to other aspects of milling operation.
28. Beware of loose clothing near the rotating parts of the milling machine.
29. Beware of loose hair near the rotating parts of the milling machine.
30. Beware of performing another operation while in close proximity to rotating parts on the milling machine.
31. Always attend to filing and deburring operations.

Always pay attention to file or deburring tools close to the cutting tool.

Files or deburring tools may catch on the cutting tool.

32. Beware of clutch (where applicable) position when jogging the spindle to different positions for gauging.
33. Beware of hands resting on clutch levers.
34. Be sure the milling machine is shut off when placing gauges on components in position.
35. Be sure the motor (on machines with clutches) is not running when using gauges on the machine.
36. Always wear protection before operating the milling machine.

Always wear the correct protection before operating the milling machine.

Never remove protection for even a short time when operating the milling machine.

Wear protective devices correctly.

Know the correct way to wear protective devices.

37. Beware of material flying from the milling machine.
38. Keep protective guards at the point of operation.

Know how to set or attach protective guards properly.

Never use the wrong protective guard.

Know how to select the proper guards.

39. When the spindle and/or the workpiece are in motion never reach over, under or around a workpiece to make an adjustment.

Never reach over, under or around a workpiece to retrieve anything.

Beware of where you leave your tools during set up.

Never reach over, under or around a workpiece to move hand tool to another position.

Never reach over, under or around a workpiece to tighten a milling part.

Never reach over, under or around a workpiece to remove swarf (chips).

40. Know the proper procedure for applying loads.

Never apply force from an awkward position.

41. Never mount a workpiece too large for the milling machine.

42. Never mount a workpiece too large for the operator to handle.

43. Use the equipment necessary for handling workpieces.

44. Never apply undue force on accessory or control lever.

45. Secure all workpieces.

46. Secure all jaws, nuts, bolts and locks.

47. Always use the correct equipment.

48. Never take cuts beyond the machine's capability.

49. Never use excessive force in polishing, filling and deburring.

50. Always use a proper hand tool to remove swarf (chips).

Never hurry to remove swarf (chips)

Beware of swarf (chips) wrapped around the cutting tool or workpiece.

51. Never change gears by moving them with your hands.

52. Beware of tools or parts falling on controls.